

GED Math Practice: Data, Statistics, and Probability

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How to Use This Packet

There are **20 problems** covering data types, tables and sampling, graphs, measures of center/spread, boxplots, probability (including conditional), and counting. Work them first without notes, then check the fully worked solutions.

1 Practice Questions

1. A survey records: eye color (blue, brown, green), height (in cm), and number of pets. Classify each variable as categorical, numerical discrete, or numerical continuous. Explain briefly.
2. In a class of 28 students, the number who bring lunch from home on a given day is recorded over 10 days as:
12, 14, 17, 12, 18, 16, 15, 11, 19, 16.
Create a frequency table for the distinct values and a relative-frequency table rounded to three decimals.
3. A company wants to estimate the average commuting time of its 900 employees. Propose a *stratified random sampling* plan and describe one possible source of bias that could still remain.
4. Commute times (in minutes) for 12 people are:
8, 12, 14, 18, 20, 22, 25, 27, 28, 35, 40, 42.
Using bin width 10 starting at 0, list the histogram bin counts.
5. A dot plot shows most data between 60 and 70 with a few values at 45 and 92. Describe the center (approximate), spread, any skewness, and outliers.
6. For category counts A: 24, B: 18, C: 8, D: 10, compute the *pie chart* central angle for each category (nearest degree).
7. For the data set 4, 7, 7, 8, 10, 12, 15, compute the mean, median, mode(s), and range. Show steps.
8. For the data set 3, 5, 6, 6, 7, 9, 12, 15, find Q_1 , Q_3 , and the interquartile range (IQR). Use the IQR rule to determine if 15 is an outlier.
9. The ages (in years) of nine participants are 18, 19, 20, 21, 25, 27, 27, 29, 35. Compute the mean absolute deviation (MAD) from the mean.
10. A five-number summary is: $\min = 30$, $Q_1 = 45$, Median = 50, $Q_3 = 60$, $\max = 78$. Sketch in words (no drawing needed) what the boxplot indicates about center and spread, and state what percentage of data lie between 45 and 60.
11. A fair six-sided die is rolled once. Find the probability of getting (a) an odd number, (b) a number greater than 4, (c) an odd number or a number greater than 4. Show formulas.
12. A bag contains 6 red, 5 blue, and 4 green marbles. Two marbles are drawn *without replacement*. Find the probability that both are blue.
13. Two cards are drawn from a standard 52-card deck *without replacement*. Find the probability that both are hearts.
14. Two independent coins are flipped three times each (so six flips total). What is the probability of getting exactly 4 heads in the six flips?
15. The two-way table summarizes 120 students by grade and whether they participate in after-school clubs.

	Participate	Do Not	Total
Grade 9	18	12	30
Grade 10	20	10	30
Grade 11–12	36	24	60
Total	74	46	120

1. Find $P(\text{Participate})$.
2. Find $P(\text{Participate} \mid \text{Grade 9})$.
3. Are ‘Participate’ and ‘Grade 9’ independent? Justify numerically.
1. A password consists of 4 characters chosen from the 26 letters (A–Z) with no repeats. How many possible passwords are there? Now suppose characters can be letters *or* digits 0–9 (36 symbols) with no repeats; how many?
2. From 10 applicants, a committee of 4 is chosen. (a) In how many ways can the committee be formed? (b) If two particular people refuse to serve together, how many committees are possible?
3. A multiple-choice quiz has 8 questions with 4 choices each. If a student guesses randomly on all questions, what is the probability of getting exactly 5 correct? (Assume independence.)
4. In a class, 40% take Art (A), 35% take Music (M), and 15% take both. Find (a) $P(A \cup M)$, (b) $P(\text{neither})$, and (c) $P(A \mid M)$.
5. A small factory inspects every 5th item off the line. Name the sampling method and describe a potential risk with this approach. Suggest one improvement.

2 Fully Worked Solutions

1. **Classification.** Eye color is **categorical** (labels). Height is **numerical continuous** (measured on a scale, can take any real value within a range). Number of pets is **numerical discrete** (counts: 0,1,2,...).
2. **Tables.** Distinct values: 11, 12, 14, 15, 16, 17, 18, 19.
Frequencies: 11:1, 12:2, 14:1, 15:1, 16:2, 17:1, 18:1, 19:1. Total $n = 10$.
Relative frequencies (to 3 d.p.): 11:0.100, 12:0.200, 14:0.100, 15:0.100, 16:0.200, 17:0.100, 18:0.100, 19:0.100.
3. **Stratified plan.** Divide employees into strata (e.g., department or work-site, or full-time vs part-time). Randomly select proportional samples from each stratum (e.g., 10% from each department). *Possible bias:* nonresponse bias (those with long commutes skip survey) or time-of-day bias (only day-shift sampled). Mitigate with reminders and varied contact times.
4. **Histogram bins (width 10 from 0).**
Bins [0,10):1 (8); [10,20):3 (12,14,18); [20,30):5 (20,22,25,27,28); [30,40):2 (35,40); [40,50):1 (42).
5. **Shape description.** Center ≈ 65 (most between 60–70). Spread roughly 45 to 92. Likely slight *right skew* due to large value at 92; values at 45 and 92 may be outliers depending on IQR rule (insufficient data to confirm but they are potential outliers).
6. **Pie angles.** Total = $24 + 18 + 8 + 10 = 60$. Angle = fraction $\times 360^\circ$.
A: $(24/60)360 = 144^\circ$; B: $(18/60)360 = 108^\circ$; C: $(8/60)360 = 48^\circ$; D: $(10/60)360 = 60^\circ$.
7. **Center/spread.**
Sum = $4 + 7 + 7 + 8 + 10 + 12 + 15 = 63$, $n = 7$. Mean $\bar{x} = 63/7 = 9$. Median is 4th value (sorted): 8. Mode: 7 (appears twice). Range = $15 - 4 = 11$.
8. **Quartiles and outliers.**
Sorted: 3, 5, 6, 6, 7, 9, 12, 15 ($n = 8$). Median = $(6 + 7)/2 = 6.5$. Lower half 3, 5, 6, 6 $\Rightarrow Q_1 = (5 + 6)/2 = 5.5$. Upper half 7, 9, 12, 15 $\Rightarrow Q_3 = (9 + 12)/2 = 10.5$. IQR = $10.5 - 5.5 = 5$. Fences: $5.5 - 1.5(5) = -2.0$ and $10.5 + 1.5(5) = 18.0$. Since $15 < 18$, **15 is not an outlier**.
9. **MAD.**
Data: 18, 19, 20, 21, 25, 27, 27, 29, 35. Sum = 221, $n = 9$. Mean $\bar{x} = 221/9 \approx 24.556$.
Absolute deviations (rounded to 3 d.p.):
 $|18 - 24.556| = 6.556$, $|19 - 24.556| = 5.556$, $|20 - 24.556| = 4.556$, $|21 - 24.556| = 3.556$,
 $|25 - 24.556| = 0.444$, $|27 - 24.556| = 2.444$, $|27 - 24.556| = 2.444$, $|29 - 24.556| = 4.444$,
 $|35 - 24.556| = 10.444$.
Sum ≈ 40.0 ; MAD $\approx 40.0/9 \approx 4.444$.
10. **Boxplot interpretation.** Median at 50 is the center of the box 45–60 (IQR = 15). Whiskers to 30 and 78 indicate full range; data are moderately spread with middle 50% between 45 and 60. About **50%** of the data lie between 45 and 60 (inside the box).
11. **Single-roll probabilities.**
(a) Odd numbers {1, 3, 5}: $3/6 = 1/2$.
(b) Greater than 4: {5, 6}: $2/6 = 1/3$.

- (c) $P(\text{odd} \cup > 4) = P(\text{odd}) + P(> 4) - P(\text{odd} \cap > 4)$. Intersection is $\{5\}$: $1/6$. So $1/2 + 1/3 - 1/6 = 2/3$.
12. **Two draws without replacement (both blue).**
First blue $= 5/15 = 1/3$. Second blue given first $= 4/14 = 2/7$. Product $= (1/3)(2/7) = 2/21 \approx 0.0952$.
13. **Two hearts without replacement.**
 $P(\text{first heart}) = 13/52 = 1/4$. Then $P(\text{second heart} \mid \text{first}) = 12/51 = 4/17$. Product $= (1/4)(4/17) = 1/17 \approx 0.0588$.
14. **Exactly 4 heads in 6 flips.**
This is binomial: $n = 6$, $p = 1/2$. $P(X = 4) = \binom{6}{4}(1/2)^6 = 15/64 \approx 0.2344$.
15. **Two-way table.**
(a) $P(\text{Participate}) = 74/120 = 0.61\bar{6}$.
(b) Within Grade 9 total 30, participate 18 $\Rightarrow P = 18/30 = 0.6$.
(c) Independence would require $P(\text{Participate} \mid \text{G9}) = P(\text{Participate})$. Since $0.6 \neq 0.61\bar{6}$, they are *not exactly* independent (difference is small but not equal).
16. **Passwords (permutations).**
Letters only, no repeats: ${}_{26}P_4 = 26 \cdot 25 \cdot 24 \cdot 23 = 358,800$.
Letters/digits (36 symbols), no repeats: ${}_{36}P_4 = 36 \cdot 35 \cdot 34 \cdot 33 = 1,413,720$.
17. **Committees.**
(a) $\binom{10}{4} = \frac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1} = 210$.
(b) Count all committees minus those containing both specific people A & B. If A and B must both be on, choose the other 2 from remaining 8: $\binom{8}{2} = 28$. So valid committees $= 210 - 28 = 182$.
18. **Binomial (guessing).**
 $X \sim \text{Bin}(n = 8, p = 1/4)$. $P(X = 5) = \binom{8}{5}(1/4)^5(3/4)^3 = 56 \cdot \frac{1}{1024} \cdot \frac{27}{64} = \frac{56 \cdot 27}{65536} = \frac{1512}{65536} \approx 0.0231$.
19. **Inclusion-exclusion and conditional.**
(a) $P(A \cup M) = 0.40 + 0.35 - 0.15 = 0.60$.
(b) $P(\text{neither}) = 1 - 0.60 = 0.40$.
(c) $P(A \mid M) = \frac{P(A \cap M)}{P(M)} = \frac{0.15}{0.35} \approx 0.4286$.
20. **Sampling method and risk.**
This is **systematic sampling** (every 5th item). Risk: if there is a periodic defect aligned with the 5-item cycle, the sample could miss or over-represent defects, causing bias. Improvement: use a **simple random sample** or vary the starting point each time (random start) and/or combine with stratified checks across different shifts/machines.